Computational Fluid Dynamics Analysis of Inlet Jet Angle and Geometry Effects on Particle Removal in a Mixed-Cell Raceway

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Abstract

Computational fluid dynamics (CFD) simulations were conducted to investigate particle transport and removal in a commercial-scale mixed-cell raceway (MCR), with emphasis on the relative influence of inlet jet designs. A three-dimensional CFD model of a two-cell MCR was developed, and Lagrangian discrete phase model (DPM) particle tracking was used to evaluate particle removal behavior under multiple configurations. Two injection designs are evaluated across five total cases within a two-cell segment of a larger commercial-scale MCR. Design 1 examines the effect of inlet jet angle as the independent variable with orifice inlets, comparing horizontal (0°) and angled (30°) injection configurations. Design 2 uses slot inlets and investigates slot length as the independent variable, testing three configurations at 100%, 70%, and 40% of the initial slot length while maintaining a constant flow rate. Particle survival functions derived from residence-time distributions were used to quantify removal performance over the first several hydraulic retention times for both jet and surface particle injection modes. The results demonstrate that Design 2 Case 2, utilizing 70% of the initial slot length, achieved the most effective particle flushing performance, retaining only 31.83% of particles at 1 HRT compared to 38.41% for the best-performing Design 1 configuration. Within Design 1, horizontal injection significantly outperformed the 30° case, which additionally produced an unfavorable exit distribution with a disproportionate fraction of particles escaping through the outlet weir rather than the bottom drains. Across all configurations, Drain 2 accounted for the largest share of particle exits, attributed to the pulling effect of the outlet weir within the reduced two-cell model. These findings highlight inlet design as a primary control on self-cleaning performance in mixed-cell raceways.

Keywords: Mixed-Cell Raceway, Computational Fluid Dynamics, Discrete Phase Model, Particle Tracking, Solids Removal

1. Introduction

Recirculating aquaculture systems (RAS) continue to expand globally as producers strive to increase fish production while reducing water use, waste discharge and environmental impact. In 2024, the number of fish raised in aquaculture surpassed the number from capture fisheries. Vital to RAS operations is the design of tanks that provide consistent hydraulic conditions, efficient solids removal and an environment that supports fish health. Traditionally, circular and rectangular tanks were used, with each offering its benefits and drawbacks. Circular tanks promote strong rotational flow and increased waste collection efficiency. Rectangular tanks maximize the usable area but have uneven mixing and increased dead zones (Wu et al., 2025).

Mixed cell raceways (MCRs) were developed to combine the self-cleaning attributes and uniform water quality conditions of circular tanks with the space efficiency and linear flow patterns of raceways, along with having manageable widths and depths. An MCR consists of a series of rotational cells designed to increase efficiency of waste collection, transport of food, minimize dead zones, maintain a uniform hydraulic condition, and improve growth performance of fish (Chun et al., 2018; Mohammadiazarm et al., 2023). Although prototype MCRs have been shown to be promising, their behavior and inlet conditions, such as the nozzle geometry and size, influence the water circulation strength and self-cleaning efficiency, leading to much variability between MCRs (Moghadam et al., 2024).

Previous analysis of MCR and RAS's have relied heavily on physical models, both at a small scale and full scale prototypes (Chun et al., 2018). While useful, it is difficult to scale a small experiment due to the size of the nozzles. In full scale models, it is difficult to understand the three-dimensional circulation patterns that govern solids transport when imaging the model from above (Ji et al., 2024). Existing computational studies have employed two-dimensional CFD (Labatut et al., 2015), which does not cover the vertical mixing near the drains which are characteristic of the cells. In the three-dimensional CFD analysis of MCRs, Chun et al. investigated replacing elevated side drains with an inlet and outlet weir located at the ends of the MCR, though the weirs did not have a significant effect on the vortices of the fluid (Chun et al., 2018).

A significant research gap exists in the quantitative evaluation of self-cleaning performance in aquaculture tanks (Wu et al., 2025). In commercial mixed-cell raceways, solids removal is governed both by how flow exits the system and by the strength of internal circulation generated at the inlet. This study examines two complementary design variables in a commercial-scale MCR. First, the inlet jet angle is varied to assess its influence on the bulk rotational flow structure and resulting particle flushing performance within the tank. Here, the angle of entry refers to the orientation of the velocity vector at the jet injection boundary, with cases that span horizontal injection and an angled configuration of 30°. Second, the geometry of the inlet is modeled as a slot and is varied by systematically reducing the length of the slot while maintaining a constant total flow rate, allowing an assessment of how the dimensions of the slot influence the self-cleaning ability of the tank without altering the overall hydraulic forcing. Together, these two design variables are evaluated across a total of five cases to identify configurations that most effectively promote particle evacuation and minimize residence time within the raceway.

Rather than relying solely on local flow metrics, self-cleaning efficiency is quantified using particle survival within the MCR as the primary performance measure, as particle survival directly captures the integrated effects of circulation, vertical mixing, near-bed transport, and drainage efficiency. Prolonged particle survival indicates ineffective solids removal and the presence of dead zones, whereas rapid removal signifies effective self-cleaning. The objective of this study is therefore to quantitatively evaluate how outflow partitioning and inlet jet geometry and velocity jointly influence particle survival in a commercial-scale MCR.

2. Materials and Methods

Self-cleaning performance in the commercial-scale MCR was evaluated using a parametric CFD study centered on particle transport and removal. Two injection designs are evaluated across five total cases within a two-cell segment of a larger commercial-scale MCR. Design 1 uses orifice inlets and examines the effect of inlet jet angle as the independent variable, comparing horizontal (0°) and angled (30°) injection configurations. Design 2 uses an alternative inlet shape and investigates this new slot length as the independent variable, testing three configurations at 100%, 70%, and 40% of the initial slot length while maintaining a constant flow rate. In both designs, the dependent variables of interest are the particle survival fraction at each hydraulic retention time interval and the particle exit distribution across the two bottom center drains and outlet weir. Two types of downlegs were used: single and double. These can be distinguished in Figure 1: a single dowleg contains one column of inlet jets, indicated by one arrow. This is also applied to a double downleg: containing two columns of inlets represented by two arrows.

For all cases, solid transport was evaluated using a Lagrangian particle-tracking approach. The dependent variable was defined as the fraction of particles introduced through the inlets that exited the computational domain as a function of objective simulation time. To enable consistent comparison across cases with differing inlet geometries and flow distributions, particle removal was additionally evaluated at the first three multiples of the hydraulic retention time (HRT). This HRT-based reporting serves as a normalization reference and does not replace objective time, but accounts for variations in turnover time arising from changes in inlet configuration and outflow partitioning. In this manner, the self-cleaning performance of the MCR was quantified through particle survival over time across all inlet configurations.

All simulations were performed using the computational fluid dynamics software ANSYS Fluent, which solves the incompressible Navier–Stokes equations over discrete control volumes subject to prescribed initial and boundary conditions. The effect of fish or their activity were not considered in this study though it is known that the presence of fish would improve solids removal due to the swimming motion mixing the water further, allowing solids to be caught by the drains. We expect our results to be conservative in our recommendations of flow rate to achieve self-cleaning.

2.1. Governing Equations

The fluid flow in the mixed-cell raceway was modeled using the incompressible Reynolds-Averaged Navier–Stokes (RANS) equations. The RANS framework decomposes the instantaneous velocity field into mean and fluctuating components, allowing the time-averaged effects of turbulence on the mean flow to be captured without resolving all turbulent scales explicitly. This approach provides a computationally efficient and robust description of a steady flow field, making it well suited for our simulation cases.

For an incompressible fluid, the RANS momentum equation is given by

$$\frac{\partial \overline{u}_i}{\partial t} + \overline{u}_j \frac{\partial \overline{u}_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial \overline{p}}{\partial x_i} + \nu \frac{\partial^2 \overline{u}_i}{\partial x_j \partial x_j} - \frac{\partial}{\partial x_j}\left(\overline{u_i' u_j'}\right) + g_i, \quad (1)$$

where $\overline{u}_i$ is the time-averaged velocity component in the ith direction, $\overline{p}$ is the mean pressure, ρ is the fluid density, ν is the kinematic viscosity, g_i is the gravitational acceleration, and $\overline{u_i' u_j'}$ represents the Reynolds stress tensor arising from velocity fluctuations.

The corresponding continuity equation for incompressible flow is

$$\frac{\partial \overline{u}_i}{\partial x_i} = 0. \quad (2)$$

The Reynolds stress term in Eq. (1) introduces additional unknowns associated with turbulent velocity fluctuations, resulting in a closure problem (Ansys Inc., 2007)

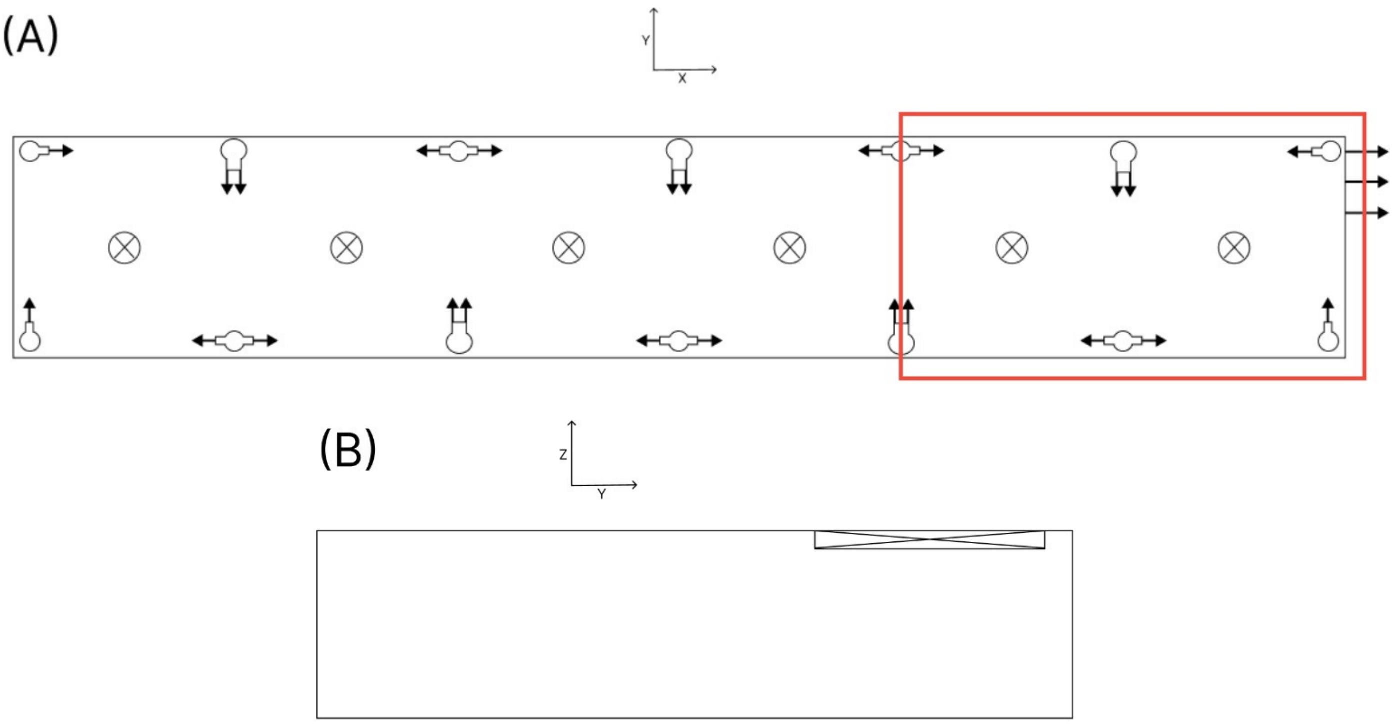

Figure 1: (A) Top view of the mixed-cell raceway model. Arrows indicate the local flow directions at the inlets and outlet. The number of inlet arrows corresponds to the number of jet columns in each downleg. The circles running accross the midline represent drains. The red box outlines the simulation region within this paper. (B) Side view showing outlet weir geometry.

To close the RANS equations, a turbulence model is required to relate the Reynolds stresses to the mean flow variables. The numerical model selected to provide this closure is described in Section 2.6.1.

2.2. Mixed Cell Raceway Designs

The mixed-cell raceway (MCR) configurations varied depending on the experimental conditions. For all cases, each MCR had overall dimensions of 5.00 m $\times$ 10.0 m $\times$ 1.67 m (16.4 ft $\times$ 32.8 ft $\times$ 5.47 ft) in width, length, and depth, respectively. The raceway was divided into two equal cells, each with dimensions of 5.00 m $\times$ 5.00 m $\times$ 1.67 m (16.4 ft $\times$ 16.4 ft $\times$ 5.47 ft). Each cell contained a centrally located circular drain with diameter 0.150 m ($\sim$ 6 in). An outlet weir was positioned near the downstream end of the raceway with dimensions of 2.00 m $\times$ 0.167 m (6.56 ft $\times$ 0.548 ft) in length and height, respectively, and located 0.167 m away from the right wall. The outlet weir represents the downstream discharge location through which treated water exits the MCR and is conveyed to the treatment system. Flow circulation within each cell was generated using vertical jet manifolds (downlegs) arranged to produce counter-rotating vortices. As shown in Figure 1, the simulated region is only a portion of the full MCR. Analyzing a two-cell MCR was done to reduce computational costs and streamline data analysis for the mentioned test cases. The conclusions gained from a two-cell MCR can bring about valuable insight for MCRs with a greater number of cells due to commonalities in the model, numerical solution, and applied boundary conditions.

2.3. Design 1: Nozzle-Based Configuration

The first configuration (Design 1) of the single-sided downleg consisted of four equally spaced 50 mm diameter nozzles oriented perpendicular to the cell sidewall, with the lowest nozzle positioned 0.333 m above the tank floor. The single-sided double downleg consisted of two vertically aligned columns of four nozzles each, separated horizontally by 10 mm while maintaining otherwise identical geometric characteristics. Double-sided downlegs were modeled as rectangular manifolds with dimensions 100 mm $\times$ 10.0 mm, featuring four equally spaced nozzles on each face oriented in opposite directions. To preserve computational efficiency, nozzles were modeled as circular inlet faces. This approach captured the dominant jet momentum and flow direction while avoiding unnecessary geometric complexity associated with explicitly modeling physical nozzle geometry.

2.4. Design 2: Slot-Based Configuration

The alternative jet manifold configuration (Design 2) consisted of downlegs with inlets modeled as rectangular slots. Each single-sided and double-sided downleg was represented as a single rectangular slot with varying width (W) and length (L), oriented in the same direction as the nozzles used in Design 1. Each slot was positioned $\frac{1.67-L}{2}$ away from both the top and bottom surfaces of the tank. To maintain constant momentum flux across all experiments, the effective slot area was constrained to equal the total nozzle area associated with a single downleg in Design 1.The single-sided double downleg was modeled as a slot with dimensions $2W \times L$, analogous to the two vertically aligned nozzle columns used in Design 1.

2.5. Simulation Cases

Experiment 1 uses two cases (0° and 45°) in order to evaluate the effects of the angle of inlet velocity on particle residence times. Experiment 2 uses three cases in order to evaluate the effects of slot dimensions on particle residence times.

2.5.1. Experiment 1: Varied Inlet Velocity Angle

For the inlet velocity comparison, the base case included an inlet velocity of 0.74 m/s at the inlet surface normal; this will be considered an angle of 0°. Setting this inlet velocity results in an HRT of 30 minutes. For outflow partitioning, the center drains were set to a loading rate of 20%, typical for physical systems. For comparison, the second case includes an inlet velocity of the same magnitude at an angle of 30° downward from the inlet surface normal.

2.5.2. Experiment 2: Varied Slot Dimensions

In this experiment, the imparted momentum flux must be kept constant. As the inlet velocity remains unchanged, the area must also remain constant. The slots within each configuration are centered between the top and bottom walls of the tank and offset from the edge of the sidewalls (where relevant) by 50 mm.

The base case includes a slot length of 1000 mm and a width of 7.855 mm. The following two cases reduce the slot length by 30% and 60%, respectively. The slot length becomes 700 mm in the second case and 400 mm in the third case. The width adjusts accordingly in order to keep the slot area constant.

2.6. CFD Implementation

2.6.1. Numerical Model Selection

Closure of the RANS equations was achieved using the k–k_l–ω transitional turbulence model as implemented in ANSYS Fluent. This three-equation eddy-viscosity model extends the standard k–ω formulation by introducing a transport equation for the laminar (pre-transitional) kinetic energy k_l, enabling the modeling of laminar, transitional, and fully turbulent flows.

The transport equation for the total turbulent kinetic energy k_T is given by

$$\frac{Dk_T}{Dt} = P_{k_T} + R + R_{\mathrm{NAT}} - \omega k_T - D_T + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\alpha_T}{\alpha_k}\right)\frac{\partial k_T}{\partial x_j}\right], \tag{3}$$

where P_{k_T} is the production of turbulent kinetic energy due to mean shear, R is the production due to bypass transition, R_{NAT} is the production due to natural (instability-driven) transition, D_T represents near-wall dissipation, ν is the molecular kinematic viscosity, and α_T is the total eddy diffusivity.

The transport equation for the laminar kinetic energy k_l is

$$\frac{Dk_l}{Dt} = P_{k_l} - R - R_{\mathrm{NAT}} - D_l + \frac{\partial}{\partial x_j}\left(\nu \frac{\partial k_l}{\partial x_j}\right), \tag{4}$$

where P_{k_l} represents the production of laminar kinetic energy by mean shear. The sink terms R and R_{NAT} govern the transfer of energy from laminar fluctuations to turbulent fluctuations through bypass and natural transition mechanisms, respectively, and D_l represents dissipation of laminar kinetic energy.

The transport equation for the inverse turbulent time scale ω is given by

$$\begin{aligned}\frac{D\omega}{Dt} &= \alpha_\omega \frac{\omega}{k_T} P_{k_T} + \gamma_\omega \frac{\omega}{k_T}\left(R + R_{\mathrm{NAT}}\right) - \beta_\omega \omega^2 \\ &\quad + \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\alpha_T}{\alpha_\omega}\right)\frac{\partial \omega}{\partial x_j}\right] + D_\omega,\end{aligned} \tag{5}$$

where γ_ω is a model constant governing the coupling between transition mechanisms and the turbulent time scale, and D_ω represents a composite near-wall damping term.

By explicitly modeling the growth and breakdown of laminar kinetic energy, the k–k_l–ω model enables realistic prediction of transitional behavior, intermittency, and near-wall damping. This makes it well suited for flows with strong shear and recirculation, such as those present in mixed-cell raceways. A more complete treatment of these governing equations can be found in Ansys Inc., 2007.

2.6.2. Boundary Conditions

Boundary conditions were established at the inlet jets (slots/orifices), two end walls, and bottom, side, and top surfaces of the raceway. In each case, the inlet jet surfaces were defined as `velocity-inlet` with the corresponding velocity value of 0.74 m/s. In the first case of Design 1 (0°), the velocity value was applied as a magnitude normal to the face direction. Due to this, all inlet jets could be grouped into one named selection. In the 30° case, each down leg of orifices was assigned to an independent named selection. The velocity components were specified to accommodate the 30° (downward) angle while maintaining the same velocity magnitude.

For the two center drains, these were assigned as `pressure-outlet` with an imposed target mass flow rate (of 4.62 kg/s) corresponding to a 20% outflow through the center drains. When the target mass flow rate is defined, Fluent adjusts pressure at the `pressure-outlet` area at every iteration in order to maintain the designated flow rate. The outlet weir was also defined as a `pressure-outlet` with 0 gauge pressure to represent the weir being open to atmospheric pressure. For the top surface, a perfect slip wall condition was assigned to ensure that the fluid remained fully contained without removing kinetic energy due to friction. The side wall surfaces were assigned `no-slip` wall conditions.

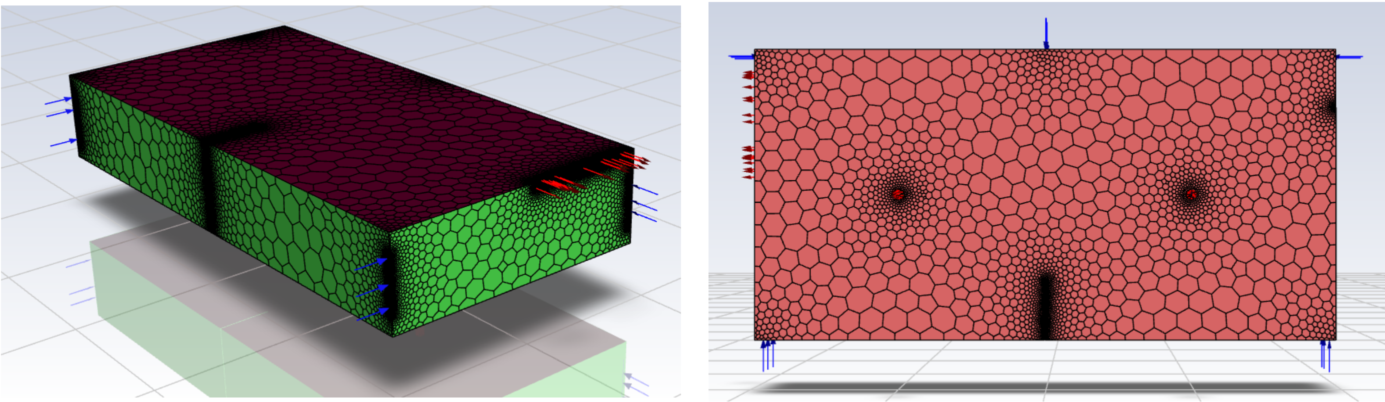

Figure 2: Representative polyhedral volume mesh of the mixed-cell raceway showing local refinement near inlet jets and the outlet weir. Blue arrows indicate velocity inlets; red arrows indicate pressure outlets.

2.6.3. Mesh Selection

The mesh was simplified where possible to reduce computational cost while preserving the dominant physics governing circulation and particle transport within the mixed-cell raceway (MCR).

As discussed before, the double-sided downlegs were modeled as rectangular cutouts within the fluid domain rather than as fully resolved cylindrical pipes. All other downlegs within Design 1 were simply modeled as a column of circular faces representing the inlet jets. For Design 2, the inlet jets were modeled as long slender rectangles. Given the small characteristic dimensions of the downlegs compared to the MCR, these geometric simplifications were not expected to meaningfully influence the resulting circulation patterns.

Surface meshing employed local refinement in regions of high velocity gradients, including the inlet jets and outlet weir, using *face sizing* controls. A target surface element size of 0.005 m was applied at the inlet jets and outlet weir regions. Curvature resolution was enforced using a *curvature normal angle* of 18°, with *proximity* scoping enabled to capture sharp geometric transitions.

The volume mesh was generated using **polyhedral elements** via automatic conversion in Fluent Meshing. Polyhedral meshes provide improved numerical accuracy per degree of freedom compared to tetrahedral meshes due to their larger number of neighboring cells and reduced numerical diffusion, enabling comparable solution fidelity to be achieved with fewer total elements. Near-wall regions were resolved using three inflation layers with a growth rate of 1.20, a transition ratio of 0.272, and *smooth transition* enabled to capture near-wall velocity gradients imposed by the no-slip boundary condition. A representative view of the mesh is shown in Fig. 2.

2.6.4. Particle Tracking

Solids transport and removal were evaluated using the Discrete Phase Model (DPM) in ANSYS Fluent, in which particles are treated as a dilute Lagrangian phase advected through a continuous Eulerian fluid field. Under the dilute loading assumption, the particle phase does not appreciably modify the carrier flow, and particle trajectories are obtained by integrating the particle equations of motion through the resolved velocity field.

Two surface particle injections were introduced in order to assess particle residence times. The first injection released particles from the top surface of the entire tank (top injection). In the second injection (jet injection), particles were released from each jet surface. Comparing these two injections allows assessment of how the initial release location and local mixing environment influence particle transport pathways and survival.

In all simulations, particles were modeled as spherical inert particles with:

- diameter: 1 mm
- density: 1050 kg/m^3
- initial velocity: 0 m/s

As the density of the particles closely matches that of water, the induced particle motion depends strongly on the fluid flow. Within the Discrete Phase Model, interaction with the continuous phase was disabled, with tracking parameters of 5 and 200,000 respectively for the step length factor and maximum number of steps. The value for maximum number of steps was chosen to ensure the tracking completion of at least 95% of the injected particles. Within each injection, particles were injected using the face normal direction and randomized starting points. The number of injected particles remained constant by setting the number of streams equal to 200. Turbulent dispersion was enabled to better encompass the range of potential particle trajectories: the number of tries was set to 10. This results in a constant value of 2000 particles released per injection.

2.7. Post-Processing

Particle residence time statistics were obtained from the Discrete Phase Model (DPM) post-processing tools in ANSYS Fluent. For each simulation case, Fluent records a residence time for every particle, defined as the elapsed time between particle injection and its exit from the computational domain. In this context, the residence time represents the duration for which a particle remains "alive" within the mixed-cell raceway prior to removal through either the center drains or the outlet weir.

Fluent outputs the particle residence time distribution in the form of a histogram, which provides a discrete approximation of the underlying probability density function (PDF). To obtain a smooth, continuous representation suitable for comparison across cases, the residence time distributions were fitted using a gamma distribution, with probability density function

$$f(t;\, k, \theta) = \frac{t^{k-1} e^{-t/\theta}}{\Gamma(k)\, \theta^k}, \tag{6}$$

where t denotes particle residence time, k is the shape parameter, θ is the scale parameter, and $\Gamma(\cdot)$ is the gamma function. The gamma distribution was selected due to its flexibility in modeling strictly positive, non-exponential processes. This functional form captures both the initial rapid removal of short-residence particles and the long-tailed behavior associated with particles trapped in recirculation zones.

After obtaining the fitted gamma distribution, the particle survival function was then defined as

$$S(t) = 1 - F(t), \tag{7}$$

where $F(t)$ is the Cumulative Distribution Function (CDF) and $S(t)$ denotes the fraction of particles remaining within the raceway at time t. The CDF gives the fraction of particles removed by time t, which is necessary for defining survival because remaining particles depend on cumulative exit history rather than instantaneous probability density, and is obtained by integrating the probability density function over time. The survival function therefore provides a direct measure of self-cleaning performance, quantifying the fraction of solids retained in the system as a function of time. We can examine this function at multiples of HRTs.

3. Results

It was observed during post-processing that the top surface injection and inlet jet surface injection configurations exhibited nearly identical behavior and trends across all evaluated metrics, including survival curves and particle exit distributions. Therefore, for brevity, the following results focus solely on the jet injection cases, as presenting both sets of data would be largely repetitive and would not provide additional meaningful insight into comparative design performance.

3.1. Particle Residence Time: Design 1 (Orifices)

The jet injection survival curves for Design 1 reveal a substantial difference in particle flushing performance between the two injection angle configurations. As shown in Figure 3, the 0° case (horizontal injection) demonstrates a markedly faster particle evacuation rate, with the majority of particles removed by approximately 2 hydraulic retention times (HRT) and the particle fraction approaching zero by 3 HRT (90 minutes). In contrast, the 30° angled injection case retains a significantly higher fraction of particles throughout the simulation, with approximately 82% and 32% of particles still present after 1 and 3 HRT, respectively. The survival curve does not approach zero until nearly 225 minutes ($\sim$ 7.5 HRT). These results indicate that horizontal jet injection promotes a fluid flow that drives particles toward the drain outlets more efficiently than the angled configuration.

3.2. Particle Residence Time: Design 2 (Slots)

The Design 2 survival curves shown in Figure 4 compare three slot length configurations:

- Case 1: initial slot length
- Case 2: 70% initial slot length
- Case 3: 40% initial slot length

Among the three cases, Case 2 (70% slot length) demonstrated the best flushing performance, achieving the lowest particle fraction remaining at each marked time interval. Case 3 (40% slot length) produced a slightly slower initial decay compared to Case 2 but converged to similar behavior at later times. Case 1 (full slot length) consistently retained the highest fraction of particles throughout the simulation. These results suggest that an intermediate slot length optimizes the rotational flow field within the tank, while the full slot length may generate flow structures that are less effective at directing particles toward the center drains.

All three cases exhibit faster overall flushing performance relative to the 30° Design 1 configuration, with particle fractions approaching zero near 3 HRT. Comparing the best-performing case from each design, Design 2 Case 2 (70% slot length) outperforms Design 1 Case 1 (0° horizontal injection) at every hydraulic retention time interval. It can be seen in Table 1, at 1 HRT, Design 2 Case 2 retains 31.83% of particles compared to 38.41% for Design 1 Case 1. At 2 HRT, Design 2 Case 2 retains only 4.59% of particles versus 7.22% for Design 1 Case 1. By 3 HRT, both configurations have evacuated the vast majority of particles, with only 0.53% and 1.09% of particles remaining for Design 2 Case 2 and Design 1 Case 1, respectively.

3.3. Particle Exit Distribution

The exit outflow percentages across all cases and designs are presented in Figure 5. A consistent trend observed across nearly all configurations is the dominance of Drain 2 as the primary particle exit pathway, accounting for approximately 53–60% of escaped particles across the Design 2 cases and the Design 1 0° case. Drain 1 exit fractions were more moderate, ranging from approximately 23–33%

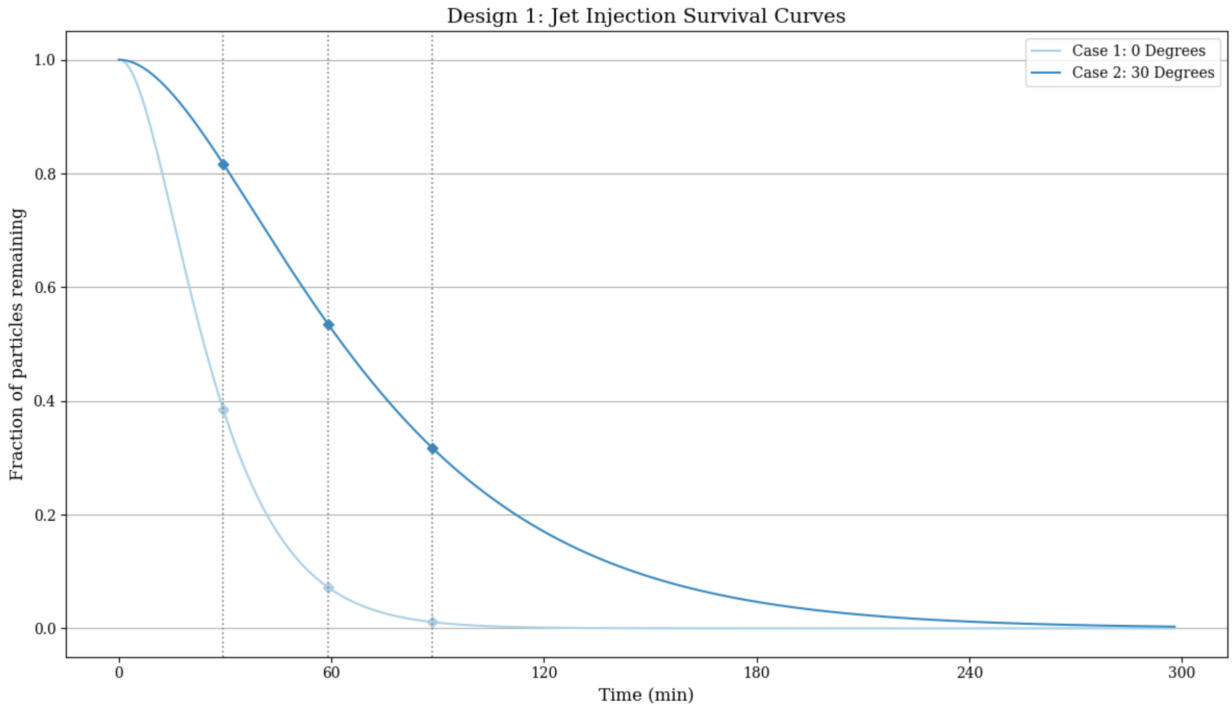

Figure 3: Particle survival functions for a jet injection for all cases of Experiment 1, with vertical markers at 1,2, and 3 multiples of the tank's HRT (30 min).

across most configurations. The consistently higher proportion of particles exiting through Drain 2 relative to Drain 1 is likely attributable to two primary factors. First, the proximity of Drain 2 to the outlet weir creates a preferential flow pathway in which the suction effect of the outlet weir draws fluid and suspended particles toward that end of the tank, effectively biasing particle trajectories toward Drain 2. Second, the present model simulates only two tank cells, representing a reduced system relative to a full-scale raceway configuration. With only two cells modeled, the hydraulic influence of the outlet weir becomes more spatially concentrated than would occur in a longer tank system.

In a configuration containing a greater number of cells, the influence of the outlet weir would be limited to near/end cells, and the exit proportions between Drain 1 and Drain 2 would be expected to converge toward more equal values, consistent with the equal loading rates imposed on each drain.

For most cases, the outlet weir captured a relatively small fraction of particles, generally remaining near 13–15%. The primary outlier in exit distribution was the Design 1 30° injection case, which exhibited a substantially different outflow profile. In this configuration, approximately 58% of particles exited through the outlet weir, while reduced fractions exited through Drain 1 ($\sim 23\%$) and Drain 2 ($\sim 19\%$). This behavior suggests that the 30° injection angle fundamentally alters the bulk flow structure within the tank, weakening the swirling motion responsible for driving settleable particles toward the bottom drains and instead allowing particles to remain suspended and exit through the surface weir. This outcome is less desirable, as solid waste removal system are typically downstream from bottom drain outflows.

4. Summary and conclusions

We employed CFD simulations to investigate the self-cleaning performance of a commercial-scale mixed-cell raceway (MCR) under variations in inlet velocity angle and inlet jet geometry (orifices vs slots). A total of five cases were evaluated across the two designs, examining the influence of injection angle in Design 1 and slot length variation in Design 2. Particular emphasis was placed on particle survival as an integrated metric of solids transport and removal, evaluated using Lagrangian particle tracking and post-processed survival functions derived from residence-time distributions. Particle residence time was quantified through survival curves tracked at one, two, and three hydraulic retention time intervals, and particle exit distributions were recorded across two center drains and an outlet weir. The major conclusions drawn from this work are as follows:

1. Design 2 Case 2, utilizing 70% of the initial slot length, achieved the most effective overall particle flushing performance, retaining only 31.83% of particles at 1 HRT and 0.53% at 3 HRT across all configurations evaluated.

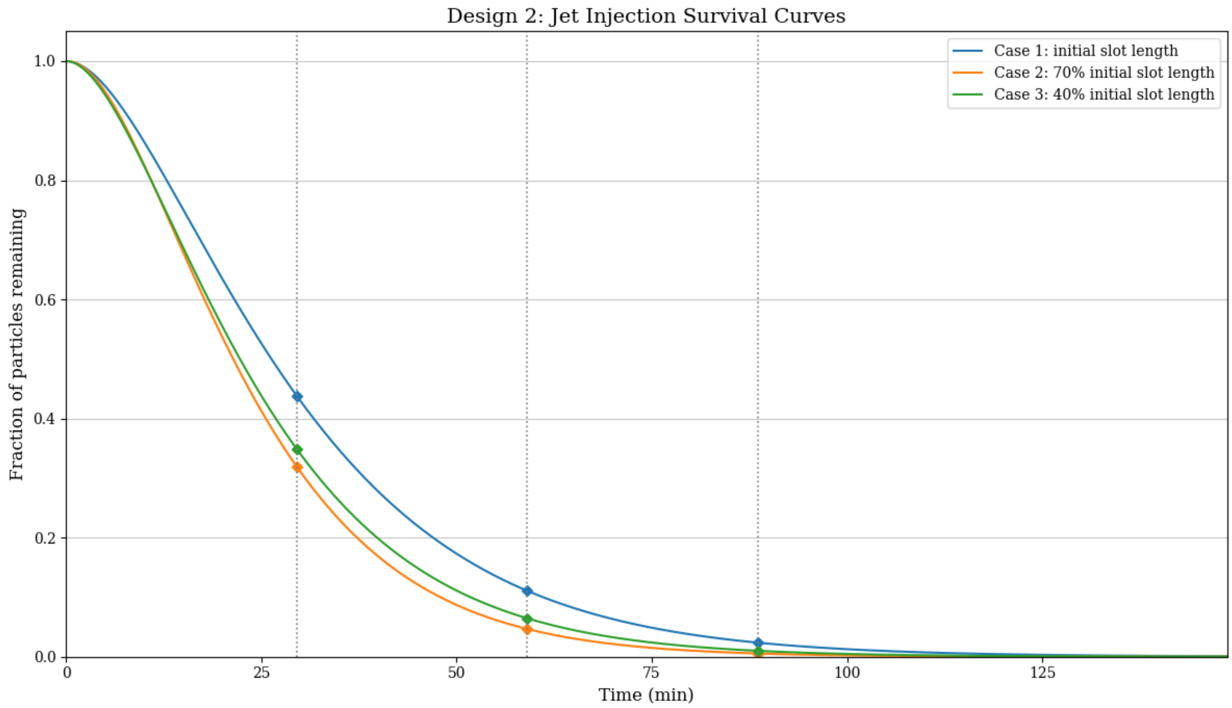

Figure 4: Particle survival functions for a jet injection for all cases of Experiment 2, with vertical markers at 1,2, and 3 multiples of HRT (30 min).

Jet Injection				
Design	Case	$S(1\,\mathrm{HRT})$	$S(2\,\mathrm{HRT})$	$S(3\,\mathrm{HRT})$
Design 1	Case 1	0.3847	0.0725	0.0011
	Case 2	0.8153	0.5339	0.3161
Design 2	Case 1	0.4384	0.1113	0.0239
	Case 2	0.3183	0.0459	0.0053
	Case 3	0.3472	0.0640	0.0099

Table 1: Fraction of particles remaining in the MCR at integer multiples of the hydraulic retention time (HRT = 30 min) for all test cases.

2. Within Design 1, the 0° horizontal injection case outperformed the 30° angled configuration at every HRT interval, retaining 38.41% and 7.22% of particles at 1 and 2 HRT respectively, compared to significantly higher fractions under the angled case.
3. The 30° injection angle produced a fundamentally different particle exit distribution, directing a disproportionately large fraction of particles through the outlet weir rather than the bottom drains. This is an undesirable outcome in aquaculture systems, where solid waste removal typically occurs downstream of bottom drains and not the outlet weir.
4. Across nearly all configurations, Drain 2 accounted for the largest share of particle exits. This is attributed to the pulling effect of the outlet weir and the limited number of tank cells modeled. In a full-scale system, this bias would be expected to diminish across a greater number of cells.
5. The top surface injection and inlet jet surface injection configurations exhibited nearly identical flushing behavior and exit distributions across all metrics, confirming that the jet injection results are representative of both configurations.

Overall, slot-based jet injection at an intermediate slot length represents the most promising design direction for efficient particle flushing in aquaculture raceways, and future work should expand the model to a greater number of tank cells, explore intermediate injection angles and slot dimensions, and pursue experimental validation of the CFD results.

Author Contributions

Ezekiel J. Evans led the computational and simulation efforts, including the development and execution of

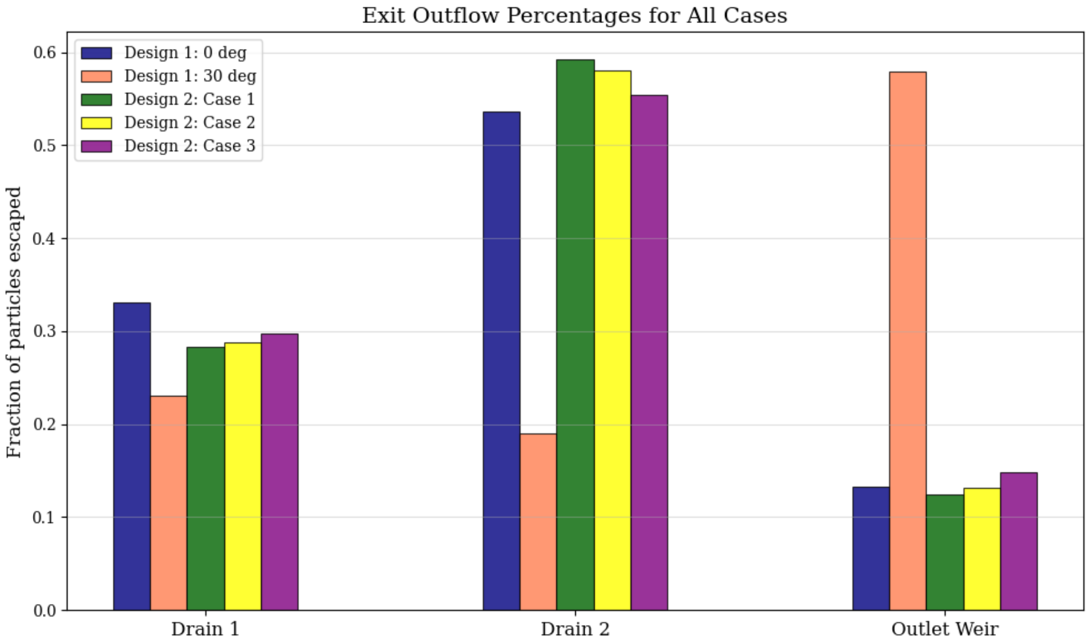

Figure 5: Particle percentages through Exit Locations for all cases.

the CFD models, analysis of results, and writing of the manuscript.

Emilio Ceballos contributed to the simulation efforts, development of CAD models, and assisted in the writing and revision of the manuscript.

Edwin Ding conducted early-stage simulations, developed the post-processing scripts used for data extraction and analysis, and assisted in the writing of the manuscript.

Natalie Sun contributed to the writing of the manuscript, led the literature review, and developed early-stage CAD models.

Michael B. Timmons served as the project advisor and provided guidance throughout the study, including insights on industry-standard practices that informed model assumptions, simulation cases, and interpretation of results.

Acknowledgments

We thank Rajesh Bhaskaran, Ph.D., Swanson Director of Engineering Simulation at the Sibley School of Mechanical & Aerospace Engineering, Cornell University, for his guidance on CFD modeling choices and for technical insight into the effective use of the ANSYS Fluent software.

Conflicts of Interest

The authors declare that they have no competing financial or personal interests that could have influenced the work reported in this paper.

References

[1] Chun, C., Vinci, B.J., Timmons, M.B., 2018. Computational fluid dynamics characterization of a novel mixed-cell raceway design. *Aquacultural Engineering* 18, 19–32. https://doi.org/10.1016/j.aquaeng.2018.02.002

[2] Ansys Inc., 2007. *FLUENT User's Guide.* Ansys Inc., Lebanon, NH, USA.

[3] Ji, X., et al., 2024. Impacts of jet inflow on hydrodynamic performance and waste removal efficiency in recirculating aquaculture system tanks based on prototype experiments. *Physics of Fluids* 36(11). https://doi.org/10.1063/5.0237057

[4] Labatut, R.A., Ebeling, J.M., Bhaskaran, R., Timmons, M.B., 2015. Exploring flow discharge strategies of a mixed-cell raceway (MCR) using 2-D computational fluid dynamics (CFD). *Aquacultural Engineering* 66, 68–77. https://doi.org/10.1016/j.aquaeng.2015.01.001

[5] Losordo, T.M., Westers, H., 1994. System carrying capacity and flow estimation. *Developments in Aquaculture and Fisheries Science* 27, 9–60.

[6] Moghadam, M., et al., 2024. Optimizing flow uniformity and velocity fields in aquaculture tanks by modifying water inlets and nozzle arrangements: A computational fluid dynamics study. *Aquacultural Engineering* 106, 102431. https://doi.org/10.1016/j.aquaeng.2024.102431

[7] Mohammadiazarm, M., et al., 2023. Growth performance and physiological responses of red tilapia reared in mixed-cell and rectangular raceways. *Aquaculture Reports* 33, 101803. https://doi.org/10.1016/j.aqrep.2023.101803

[8] Timmons, M.B., Ebeling, J.M., 2013. *Recirculating Aquaculture*, 3rd ed. Ithaca Publishing Company, Ithaca, NY.

[9] Walters, D.K., Cokljat, D., 2008. A three-equation eddy-viscosity model for Reynolds-averaged Navier–Stokes simulations of transitional flow. *Journal of Fluids Engineering* 130(12), 121401. https://doi.org/10.1115/1.2979230

[10] Wu, Y., et al., 2025. Recent advances in the hydrodynamic characteristics of industrial recirculating aquaculture systems and their interactions with fish. *Sustainability* 17(17), 7946. https://doi.org/10.3390/su17177946

[11] Timmons, M.B., Losordo, T.M. (Eds.), 1994. *Aquaculture Reuse Systems: Engineering Design and Management*. Developments in Aquaculture and Fisheries Science, Vol. 27. Elsevier Science Ltd., New York, pp. 9–60.